RALPH & TOMMY ARCHITECTURE · 1 OF 3 · THE WHOLE FLOW

How a deal flows: each step and the method behind it

Ralph is the buyer's AI agent, Tommy the seller's. The AI decides what to offer; code decides what's allowed.

step · decision · vault (secret store) · sub-process · end
code · Claude · vault · person · secret in, or a word out
Small italic text = the method's name in the code

SELLER
BUYER
CLAUDE AGENTS
CODE (FLEEK)
VAULT
PEOPLE

List lots
list price; haggle or firm price; final yes: Tommy or me
floor → vault only

lots

Brief Ralph
wants + a note; a price in the note is refused
top price → vault · check_free_text

wants

Filter lots
category, brand, grade, in stock, bundle size
SQL · explain.py

fits

Vault
floor · top price · budget · margin key
vault/ · no AI

Could any price work?
yes / no only

yes

Lot dropped

20 answers a lot a day · top price rounded down to 5% · flag 5+ accounts asking only that lot
match gate · gate.py

Rank by formula
fit 50 · price 25 · rating 15 · dispatch 10
score.py

top 5

Ralph re-ranks top 5
one reason each; public facts only
strict tool schema · each lot once · error → formula order

Open on lot #1
3 tries per lot; 24 h wait after a fail
open_negotiation

no: walked away or expired → next lot

Haggle loop
Ralph ⇄ Tommy take turns; one move each

Referee checks every card
outside the sender's limit → never sent
take_step · detail on page 2

word · yes/no

Vault answers
one band word a turn; at accept: inside both limits? Floors carry a secret margin (≤8%)
bands.py · can_accept · privacy.py

Pause · hand over · stop
either side, any time
controls.py

Agreed?
vault checked both limits

Person needed?
approval rules

yes

Monitors
flag probing, returning, crowded lots
monitors.py

People approve
buyer or seller if they chose; account manager: first 3 deals, £1,500+ delivered, flagged buyer, under 60% of list
approval.py

Deal closed
Fleek's deal strip: commission, fees, AI cost, time
append-only event log

all said yes

each side

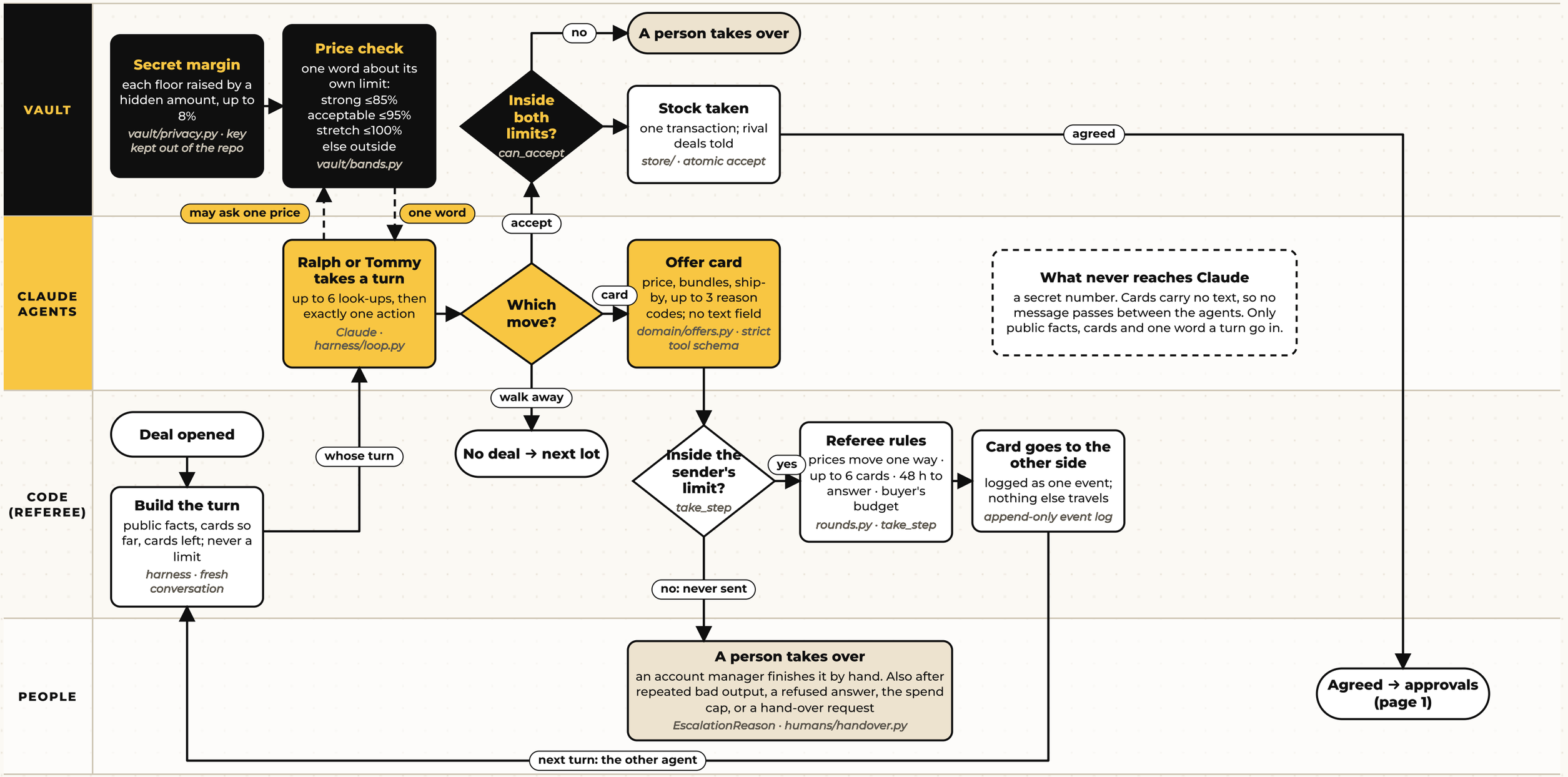
Buyer's receipt
pays delivered; under top price by
own numbers only

Seller's receipt
receives after 15%; above floor by
own numbers only

Sources (github.com/code-mohanprakash/ralph-and-tommy): market.toml [matching] 50/25/15/10, [match_gate] 20 answers a lot a day, 5% steps, 5 accounts, [negotiation] 3 tries, 24 h, [approval] £1,500 delivered, first 3 deals, under 60% of list, [privacy] margin up to 8%, [bands] 85/95%; domain/fees.py (15% commission); live/forms.py check_free_text; matching/; vault/gate.py, bands.py, checks.py, privacy.py; agents/rerank.py; orchestrator/opening.py, run.py (take_step), approval.py, controls.py; humans/monitors.py. The secret margin and match gate are off by default in market.toml and on in the live demo (cli.py).

One turn at a time: Claude proposes, code disposes

Each turn, one agent may check one price, then makes exactly one move. Code checks it before the other side sees anything.



EXAMPLE: ONE REAL LIVE RUN (CLAUDE HAIKU 5.5, LIVE DEMO)

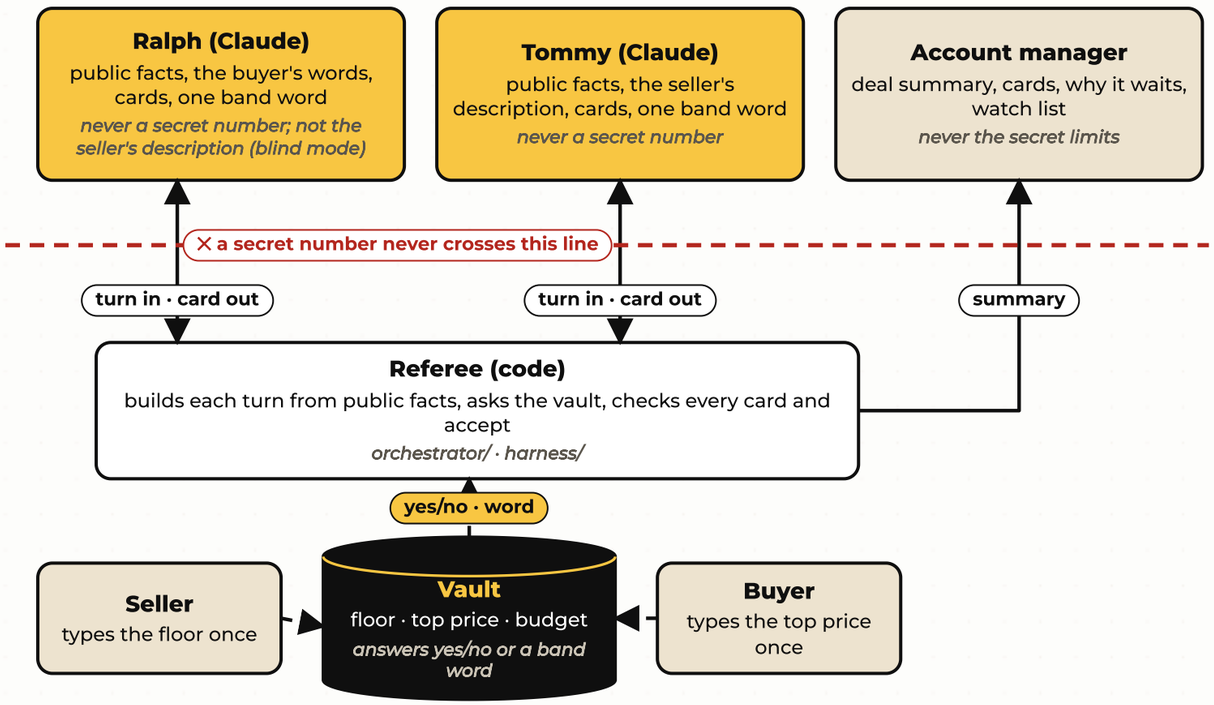
Cards: Ralph £9.00 → Tommy £14.00 → Ralph £11.50 → Tommy £12.75 → Ralph £12.00 → Tommy accepts → vault: inside both limits → buyer and account manager approve → closed at **£12.00** a piece, 100 pieces.

Price checks: 6 (strong, acceptable, acceptable, outside, stretch, stretch) · **5 cards** · 0 cards stopped by the vault · 29.9 s agent time · 22 AI calls · AI cost 0.649¢ (US). Tommy's check on £11.50 said **outside** though the seller's own floor allows £11.00: the secret margin at work.

Sources: harness/loop.py (up to 6 look-ups, one action), harness/tools.py (one price check a turn), vault/bands.py with market.toml [bands] 85/95%, vault/privacy.py (margin up to 8%, key from FLEEK_VAULT_KEY), vault/checks.py can_accept, domain/offers.py (card fields, up to 3 reason codes, no text), orchestrator/run.py take_step, orchestrator/rounds.py (one-way prices, market.toml max_rounds 6, answer_hours 48), store/ (atomic accept). **Run:** Mohan's recorded live demo (screens and screen recording). The £11.00 is £9.35 ÷ 0.85 (domain/fees.py).

Who sees what, and which folder does which job

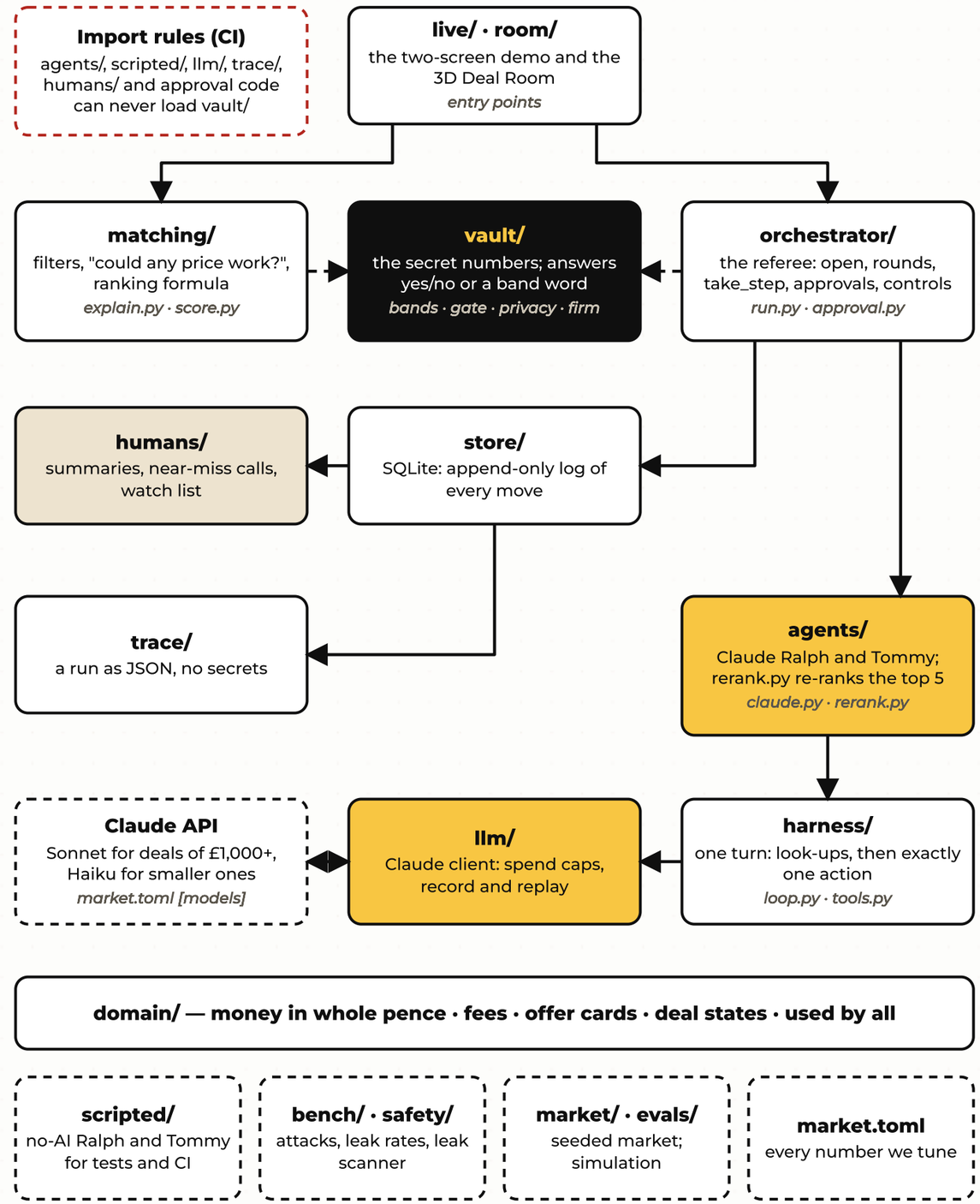
A · Who sees what



Methods at a glance

STEP	METHOD, IN PLAIN WORDS	IN THE CODE
Note check	A price typed in the buyer's note to Ralph is refused	*check_free_text*
Filters	Category, brand, grade, in stock, bundle size; each drop keeps its reason	*explain.py*
Could a price work?	Yes/no per lot; 20 answers a lot a day; top price rounded down to 5% steps; flag when 5+ accounts ask about only that lot	*gate.py*
Ranking	Fit 50 · price 25 · rating 15 · dispatch 10; fit from a local embedding model	*score.py*
Re-rank	Claude re-orders the top 5, a reason each; every lot once, else formula order	*rerank.py*
Price check	One word against its own limit: ≤85% strong, ≤95% acceptable, ≤100% stretch, else outside	*bands.py*
Secret margin	Each floor raised by a hidden amount, up to 8%	*privacy.py*
Offer card	Price, bundles, ship-by, up to 3 reason codes; no text field	*offers.py*
Referee	A card past its sender's limit is never sent; prices move one way; 6 cards; 48 h	*take_step*
Accept	Price inside both limits; stock taken in one transaction	*can_accept*
Approvals	First 3 deals, £1,500+ delivered, flagged buyer, under 60% of list, or the side chose "ask me first"	*approval.py*

B · Folder map (src/fleek_agents/)



Sources: docs/ARCHITECTURE.md ("Who knows what", "Where things live"), docs/HOW_IT_WORKS.md §1–2, pyproject.toml import-linter contracts, package imports in src/fleek_agents/ (arrows = which folder calls which), market.toml [models] (£1,000+ deals get Sonnet, smaller ones Haiku), harness/reader.py (blind mode, on in the live demo), and the files named in the table.